

Eli Pregerson

eli.pregerson@stonybrook.edu | github.com/Eli-Pregerson | <https://dblp.org/pid/351/1082.html>

Education

Stony Brook University - Stony Brook, NY
Masters of Science, Computer Science

August 2025 - Present

Harvey Mudd College - Claremont, CA
Bachelor of Science, Computer Science and Mathematics

May 2024

- National Merit Scholarship Recipient
- GPA 3.744
- Dean's List (Spring '21 - Spring '24)

Selected Coursework:

- Math: Graph Theory, Combinatorics, Single and Multivariable Calculus, Linear Algebra, Differential Equations, Discrete Mathematics, Combinatorics, Probability and Statistics, Real Analysis, Abstract Algebra, Representation Theory, Complex Analysis
- Computer Science: Algorithms(Fall '23, Spring '24 Tutor), Programming Languages, Machine Learning, Applied Logic/Automated Reasoning, Computability and Logic (Fall '22 Tutor), Natural Language Processing, Data Structures and Program Development, Principles of Computer Science (Fall '21 Tutor), Intro to CS

Awards

CRA Outstanding Undergraduate Research Award Finalist

December 2023

Description: Every year each college may nominate a number of students based on the size of the school for this national award. The title of finalist puts me in the bracket of top 9-24 students nominated that year.

Don Chamberlain Computer Science Research Award

May 2024

Description: This award is given to Harvey Mudd graduates who are judged by the faculty to have successfully completed a significant piece of computer science research.

Research

ALPAQA, Quantifying the Difficulty of Automated Testing

May - Aug 2021

Mentor: Professor Lucas Bang at Harvey Mudd College

Description: The purpose of this project was to determine the viability of a bug prediction neural network that used novel metrics developed by Prof. Bang and members of the VLab at UCSB. Our goal was to use an existing dataset of github commit messages as a proxy for the existence of bugs, and then train a neural net to predict bugs using APC and the VLab code complexity metric PReach. While working on this project I modified the outputs of the APC tool so that they would be more conducive to machine learning, removed unnecessary assumptions within the APC tool so that it could run on a much broader set of java files, and analyzed the data. At the end of this process we determined that with the dataset we had, we could not conclude anything about APC being able to predict bugs.

ALPAQA, Recursive APC

June 2021 - May 2023

Mentor: Professor Lucas Bang at Harvey Mudd College

Description: The purpose of this project was to design and implement an algorithm for accurate computation of APC for recursive functions. Previously APC had been calculated using the math of regular languages and control flow graphs, but recursive calls require the math of context free grammars to

properly match function call to function return. Once the code flow is abstracted as a context free grammar, the Chomsky-Schützenberger enumeration theorem can be used to get a generating function for the number of paths of different lengths. Going from the generating function to a closed form solution is a complex process that is actively being optimized.

Resulting Publications and Talk:

E. Pregerson, "Path Complexity of Recursive Functions," 2023 IEEE/ACM 45th International Conference on Software Engineering: Companion Proceedings (ICSE-Companion), Melbourne, Australia, 2023, pp. 308-310, doi: 10.1109/ICSE-Companion58688.2023.00083.

E. Pregerson, S. Cullen-Baratloo, D. Chen, D. Lam, M. Szostak and L. Bang, "Formalizing Path Explosion for Recursive Functions via Asymptotic Path Complexity," 2023 IEEE/ACM 11th International Conference on Formal Methods in Software Engineering (FormaliSE), Melbourne, Australia, 2023, pp. 76-85, doi: 10.1109/FormaliSE58978.2023.00016.

S. Dissem, E. Pregerson, A. Bhargava, J. Cordova and L. Bang, "Path Complexity Correlates with Source Code Comprehension Effort Indicators," 2023 IEEE/ACM 31st International Conference on Program Comprehension (ICPC), Melbourne, Australia, 2023, pp. 266-274, doi: 10.1109/ICPC58990.2023.00041.

Plateau 2021 - Path Complexity Predicts Code Comprehension

Orrison Lab, Computational Representation Theory

Jan 2023 - May 2023

Mentor: Professor Michael Orrison at Harvey Mudd College

Description: We designed algorithms for the computation of subranking and subset summary statistics in a way that maximized sharing of intermediary values and minimized the number of addition operations. We then formalized those algorithms in the language of group frames.

PRiME, Trace Ideals over semigroup rings

Jun 2023 - December 2023

Mentor: Professor Haydee Lindo at Pomona College and Harvey Mudd College

Description: The goal of this project is to categorize the structure of trace ideals over numerical semigroup rings. Trace ideals are powerful algebraic tools that can be used to classify rings and modules. Numerical semigroups rings are a pervasive type of ring that pop up in many areas of mathematics like algebraic geometry and number theory and they have a number of properties that make them a convenient object for study with trace ideals.

Resulting Talk: MAA MathFest 2023 Research in Motion Poster Session - Trace Ideals over numerical semigroup rings.

ALPAQA, Search synthesis anamorphisms and symbolic execution

Aug 2023 - Present

Mentor: Professor Lucas Bang at Harvey Mudd College

Description: The goal of this project is to formalize an existing search synthesis algorithm in terms of anamorphism and catamorphism. The algorithm builds a decision tree based on information gain after training on a dataset, which can be formalized as an anamorphism. Then for a particular element the algorithm takes the decision tree and collapses it down into a prediction, which can be formalized as a catamorphism. The input to this search synthesis is a program written in a toy language similar to a simplified c++. As part of this project I created a symbolic execution engine for this language that is also an anamorphism and can provide the search problem specification to the search synthesis algorithm.

ALPAQA, Asymptotic Path Complexity Journal Publication

Aug 2023 - Present

Mentor: Professor Lucas Bang at Harvey Mudd College

Description: One all encompassing publication of APC results will make the tool more accessible and running larger experiments will demonstrate its versatility. My contributions to this project are making the tool as simple and intuitive as possible and making the replication of experiments take as few commands as possible.

Additional Projects

MuddSub, Harvey Mudd Robosub

Sep 2020 - Jan 2022

Description: MuddSub is a student run club that builds and programs robots for the annual Robosub competition. While on the team, I worked on the controls, reinforcement learning, simulation, and power distribution teams, and I became the lead of the controls team. Most of the work I did was for the controls team, where I was responsible for taking in instructions from the Navigation team on where the robot should go, and then figuring out how much power should be sent to each thruster.

Factor, AARCH64 Compiler

Sep 2023 - May 2024

Mentor: Professor Christopher Stone

Description: The goal of this project is to produce an AARCH64 compiler backend for the concatenative stack based programming language Factor. Compiling factor into native AARCH64 machine instructions would allow for factor to be run faster on apple silicon and would make it possible to run on arm chip machines that do not have the ability to simulate x86 instructions.

Other Relevant Employment

BEAM, TA and Camp Counselor

July 2024, 2025

Description: Bridge to Enter Advanced Mathematics is a math mentorship program that aims to bring more people from under represented communities into STEM fields. As part of this program BEAM kids go to a sleep away math camp the summer after 7th grade. In this role I tutored and taught math, but I also ran normal summer camp activities.